

Stochastic Processes, Long Quiz, 2026 Fall

Solution and Grading

- Duration: 60 minutes
 - Weight: 10% of final grade
 - Closed material, No calculator
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- Write legibly.
 - Justification is necessary unless stated otherwise.

#1. Let $X \sim U(0, 4)$. Evaluate $\mathbb{E}[\min(X, 3)]$. [2.5 pt]

Since $X \sim U(0, 4)$, $f(x) = \frac{1}{4}$ for $0 \leq x \leq 4$. Therefore,

$$\begin{aligned}\mathbb{E}[\min(X, 3)] &= \int_0^3 x \frac{1}{4} dx + \int_3^4 3 \frac{1}{4} dx \\ &= \frac{1}{4} \left[\frac{x^2}{2} \right]_0^3 + \frac{3}{4} [x]_3^4 = \frac{9}{8} + \frac{3}{4} = \frac{15}{8}.\end{aligned}$$

Difficulty: Easy

Amount of work: 25%

#2. Let $X \sim \text{Poi}(3)$ and $Y = \min(X, 2)$. Find $\mathbb{E}[Y]$ and $\text{Var}(Y)$. [2.5 pt]

Since $Y = 0$ if $X = 0$, $Y = 1$ if $X = 1$, and $Y = 2$ if $X \geq 2$,
 $\mathbb{P}(Y = 0) = e^{-3}$, $\mathbb{P}(Y = 1) = 3e^{-3}$, and $\mathbb{P}(Y = 2) = 1 - 4e^{-3}$.

Thus,

$$\mathbb{E}[Y] = 3e^{-3} + 2(1 - 4e^{-3}) = 2 - 5e^{-3}.$$

Also, $\mathbb{E}[Y^2] = 3e^{-3} + 4(1 - 4e^{-3}) = 4 - 13e^{-3}$. Therefore,

$$\text{Var}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = 4 - 13e^{-3} - (2 - 5e^{-3})^2 = 7e^{-3} - 25e^{-6}.$$

Difficulty: Easy

Amount of work: 25%

#3. Let $P = \begin{bmatrix} 0.75 & 0.25 \\ 0.50 & 0.50 \end{bmatrix}$.

(a) Find $\boldsymbol{\pi} = (\pi_1 \ \pi_2)$ satisfying $\boldsymbol{\pi}P = \boldsymbol{\pi}$ and $\pi_1 + \pi_2 = 1$. [2.5 pt]

(b) Find $\boldsymbol{\pi}P^2$. [2.5 pt]

(a) From $\boldsymbol{\pi}P = \boldsymbol{\pi}$, $0.75\pi_1 + 0.50\pi_2 = \pi_1$, which gives $\pi_1 = 2\pi_2$. Together with $\pi_1 + \pi_2 = 1$, we obtain

$$\boldsymbol{\pi} = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \end{pmatrix}.$$

(b) First,

$$P^2 = \begin{bmatrix} 0.75 & 0.25 \\ 0.50 & 0.50 \end{bmatrix}^2 = \begin{bmatrix} 11/16 & 5/16 \\ 5/8 & 3/8 \end{bmatrix}.$$

Therefore,

$$\boldsymbol{\pi}P^2 = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \end{pmatrix} \begin{bmatrix} 11/16 & 5/16 \\ 5/8 & 3/8 \end{bmatrix} = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \end{pmatrix}.$$

Difficulty: Easy

Amount of work: 50%